

Introduction and Mathematical Foundations

Q1. Discuss the major goals of AI—perception, reasoning, planning, communication, creativity, learning, and control/manipulation—and explain how Generative AI is contributing to these goals.

Q2. Explain the major capabilities of Generative AI with its major applications. Explain how the scaling of Generative AI models depends on data, model parameters, computational resources, memory, and distributed training. Discuss why scaling becomes increasingly challenging as model size increases.

Q3. Differentiate between discriminative and generative models in terms of the probability distributions they learn, their objectives, and their applications. Explain how a generative model can also perform a discriminative task using the relationship between joint and conditional probability distributions.

Q4. Explain the major challenges in deep generative modeling, particularly for high-dimensional data such as images. Explain how these challenges affect the ability to generate realistic samples and perform discriminative tasks.

Q5. Explain the fundamental rules of probability, including joint probability, conditional probability, the sum rule, and the product rule. Derive Bayes' theorem from the product rule and explain the roles of prior, likelihood, posterior, and normalization constant.

Q6. Explain the Gaussian distribution and its importance in probability theory. For a dataset of independent and identically distributed (i.i.d.) observations, explain how Maximum Likelihood Estimation (MLE) can be used to estimate the mean and variance of a Gaussian distribution.

Q7. Explain the fundamental concepts of information theory. Define self-information, Shannon entropy, KL divergence, and cross-entropy. Discuss the properties of KL divergence and explain the relationship between cross-entropy and KL divergence.

Q8. Consider a medical diagnosis problem in which an X-ray image of a patient is analyzed to determine whether the patient has cancer (C_1) or no cancer (C_2). The following information is available:

Class	Prior Probability	Likelihood of X-ray Result
Cancer (C_1)	0.10	0.80
No Cancer (C_2)	0.90	0.20

Using Bayes' theorem, calculate the posterior probability of each class given the observed X-ray result. Based on the posterior probabilities, determine the appropriate classification decision. Explain the roles of prior probability, likelihood, posterior probability, and normalization constant in the decision-making process.

Deep Generative Architectures

Q1. Differentiate between discriminative modeling $P_\theta(Y|X)$, generative modeling $P_\theta(X)$, and conditioned generative modeling $P_\theta(X|Y)$. Explain their objectives with suitable examples.

Consider A classifier produces the following probabilities for three classes for a given input image:

Class	Probability
Cat	0.10
Dog	0.75
Bird	0.15

Determine the predicted class and calculate the negative log-likelihood loss if the actual class is Dog.

Q2. Discuss the importance of sampling, interpolation, and disentanglement in exploring the learned manifold of a deep generative model. Illustrate each concept with a suitable example.

Q3. Compare GANs, VAEs, and Denoising Diffusion Models in terms of their architecture, training mechanism, sampling process, and ability to generate new data.

Q4. Explain the architecture and working of a Variational Autoencoder (VAE). Discuss how the encoder predicts the mean and covariance of a multivariate normal distribution, how the decoder reconstructs the input, and why the latent distribution is encouraged to follow $N(0, I)$.

Q5. Explain why direct sampling from $N(\mu(X), \Sigma(X))$ creates a problem during backpropagation in a VAE. Explain the **reparameterization trick** and show how it enables gradient-based training.

Q6. Consider a one-dimensional latent variable with

$$\mu=2, \sigma=0.5, \epsilon=-1.2,$$

where $\epsilon \sim N(0, 1)$. Using

$$z = \mu + \sigma \epsilon,$$

calculate the latent variable z . If the decoder maps this value according to $x' = 3z + 1$, calculate the generated output x' .

MLE and Latent Variable Model

Q1. Explain the motivation for using **latent variable models** in deep generative modeling. Consider an image dataset in which factors such as gender, eye color, hair color, and pose are not explicitly annotated.

(a) Distinguish between observed variables x and latent variables z .

(b) Explain how latent variables can represent high-level factors of variation.

(c) Describe the generative model $p(x|z)$ and explain why it may be simpler than directly

modeling $p(x)$.

(d) Discuss how $p(z|x)$ can be used for unsupervised representation learning.

Q2. For a latent variable model, the log-likelihood is

$$\log p_{\theta}(x) = \log \sum_z p_{\theta}(x, z).$$

(a) Explain why Jensen's inequality is applicable to this expression.

(b) Introduce an arbitrary probability distribution $q(z)$ and rewrite the marginal likelihood in terms of $q(z)$.

(c) Apply Jensen's inequality to derive the Evidence Lower Bound (ELBO).

(d) Explain why the resulting quantity is called a *lower bound* on the log-likelihood.

Q3. For a variational distribution $q(z)$, the relationship between log-likelihood, ELBO, and KL divergence is

$$\log p(x; \theta) = \mathcal{L}(x; \theta, \phi) + D_{KL}(q(z; \phi) || p(z|x; \theta))$$

(a) Explain why $D_{KL}(q || p) \geq 0$ implies that ELBO is a lower bound.

(b) Derive the above relationship between log-likelihood, ELBO, and KL divergence.

(c) State the condition under which the ELBO becomes equal to the true log-likelihood.

(d) Explain what happens to the ELBO as $q(z; \phi)$ becomes a better approximation to the true posterior.

Q4. Suppose a latent variable z has **4 possible configurations** and the joint probabilities for a given observation x are:

$$p(x, z_1) = 0.10, \quad p(x, z_2) = 0.25, \quad p(x, z_3) = 0.05, \quad p(x, z_4) = 0.20.$$

(a) Calculate the marginal likelihood $p(x)$.

(b) Calculate $p(z_i|x)$ for each latent configuration.

(c) Identify the most probable latent configuration based on the posterior.

(d) Explain how this posterior inference can be interpreted in the context of clustering or representation learning.

Generative Adversarial Network

Q1. Explain the architecture and working principle of a **Generative Adversarial Network (GAN)**. Describe the roles of the **generator G_{θ}** and **discriminator D_{ϕ}** and explain how they participate in the two-player minimax game. Formulate the GAN objective function and explain the purpose of each term.

Q2. During discriminator training, suppose the discriminator outputs:

- $D(x)=0.9$ for a real sample

- $D(G(z))=0.2$ for a generated sample.

Using the binary cross-entropy discriminator loss,

$$L_D = -[\log D(x) + \log(1 - D(G(z)))]$$

calculate L_D . Interpret the result.

Q3. Consider for a particular minibatch, suppose the discriminator outputs the following probabilities:

Sample	Type	D(x)
x1	Real	0.9
x2	Real	0.8
G(z1)	Fake	0.2
G(z2)	Fake	0.1

Using

$$V(G, D) = \frac{1}{2} \sum_{i=1}^2 \log D(x_i) + \frac{1}{2} \sum_{i=1}^2 \log(1 - D(G(z_i))),$$

calculate the discriminator objective value $V(G, D)$ using natural logarithms. Determine whether the discriminator is performing well based on its outputs.

Q4. During one GAN training iteration, the generator produces four fake samples. The discriminator outputs are

$$D(G(z_1))=0.1, D(G(z_2))=0.3, D(G(z_3))=0.7, D(G(z_4))=0.5.$$

Using the generator objective given in the lecture,

$$L_G = \frac{1}{4} \sum_{i=1}^4 \log(1 - D(G(z_i))),$$

(a) Calculate the generator loss.

(b) Explain what the magnitude of the loss indicates about the discriminator's ability to distinguish the generated samples.

(c) If the discriminator outputs become

$$[0.6, 0.7, 0.8, 0.9],$$

calculate the new generator loss and compare the two cases.

Q5. Discuss the major practical challenges associated with training GANs. Explain unstable optimization and mode collapse with suitable examples. In your answer, describe why generator and discriminator losses may oscillate during training and explain how mode collapse affects the diversity of generated samples. Also discuss the empirical approaches used to mitigate mode collapse.

Q6. Consider a GAN with an optimal discriminator D_G^* . Show that the value function can be expressed in terms of the **Jensen–Shannon Divergence (JSD)** between p_{data} and p_G :

$$V(G, D_G^*) = 2D_{\text{JSD}}(p_{\text{data}}, p_G) - \log 4.$$

Define JSD and explain its important properties. What is the condition on p_G at the optimum of the GAN objective, and what is the corresponding value of the objective function?

Q7. Explain why evaluating generative models is challenging. Discuss how the evaluation strategy depends on the intended task. Explain the evaluation requirements for:

1. Density estimation
2. Sampling/generation
3. Latent representation learning
4. Downstream applications such as semi-supervised learning and image translation.

Q8. Explain the **Inception Score (IS)** used for evaluating the quality of generated samples. Discuss why a good generative model should simultaneously exhibit **high sharpness and high diversity**, relating to the classifier's predictive distribution $c(y|x)$ and marginal distribution $c(y)$

Q9. Explain the **Frechet Inception Distance (FID)** and its role in evaluating generative models. Suppose the feature representations of generated samples and test samples are modeled as multivariate Gaussians with parameters (μ_G, Σ_G) and (μ_T, Σ_T) , respectively. Write the FID formulation as:

$$FID = \|\mu_T - \mu_G\|^2 + \text{Tr} \left(\Sigma_T + \Sigma_G - 2(\Sigma_T \Sigma_G)^{1/2} \right).$$

Explain each term and describe the interpretation of a **low FID value**. Also explain how FID differs conceptually from the Inception Score.

Variants of GAN

1. Explain the architecture and working principle of a Conditional GAN (cGAN). How does conditioning on class information modify the conventional GAN framework? Explain how cGANs can be used for text-to-image synthesis.

2. Explain the motivation behind **BiGAN (Bidirectional GAN)**. Describe how the generator G and encoder E learn the joint distributions of (X, Z) , and explain the condition under which E and G become approximately inverse mappings.

3. Consider two semantically related image domains $X_{A \rightarrow B}$ and $X_{B \rightarrow A}$, for which **corresponding image pairs are not available**.

(a) Explain how **CoGAN (Coupled Generative Adversarial Networks)** learns the joint distribution of the two domains without correspondence supervision.

(b) Explain the role of **weight sharing** in CoGAN. Why are different layers shared between the two networks, and how does this model inductive bias help in learning common high-level semantics? Also discuss one important limitation of CoGAN.

4. Two image domains A and B contain **unpaired training images**, and the mapping between the domains is unknown.

(a) Explain the architecture and working principle of **CycleGAN** for unpaired image-to-image translation. Define the two mappings $G_{A \rightarrow B}$ and $G_{B \rightarrow A}$, and explain how adversarial loss is used.

(b) Explain the importance of **cycle-consistency loss** and **identity loss** in CycleGAN. Discuss why cycle consistency is necessary and describe a limitation of relying only on cycle-consistency loss.

5. Explain linear interpolation in the latent space of a GAN. Given two latent vectors z_1 and z_2 , derive the interpolated latent vector for a step size $\alpha \in [0,1]$, and explain how the generator produces the corresponding sequence of images. Explain the concept of **steerability of GANs**. How can transformations such as zooming, shifting, brightness adjustment, and rotation in image space be represented as transformations in latent space?

Q6. Explain the motivation for introducing **Bidirectional Generative Adversarial Networks (BiGANs)**. Compare BiGAN with a conventional GAN in terms of **latent-variable inference**.

Describe the roles of the **generator G**, **encoder E**, and **discriminator D**. Explain how the discriminator distinguishes between the joint samples

$$(z, G(z))$$

and

$$(E(x), x).$$

How can a trained BiGAN be used both for **generating new samples** and **inferring latent representations**?

Q7. Explain the problem of **image-to-image translation between two domains X and Y** when paired training examples are unavailable.

Describe the architecture of **CycleGAN**, including the two generators

$$G: X \rightarrow Y, F: Y \rightarrow X$$

and the two discriminators D_Y and D_X . Explain how adversarial training is performed in both directions.

Q8. Explain the concept of **cycle consistency** in CycleGAN. Derive and explain the complete CycleGAN objective:

$$\min_{F, G, D_X, D_Y} \mathcal{L}_{GAN}(G, D_Y, X, Y) + \mathcal{L}_{GAN}(F, D_X, X, Y) \\ + \lambda (E_X[\|F(G(X)) - X\|_1] + E_Y[\|G(F(Y)) - Y\|_1]).$$

Explain the role of the **cycle-consistency term**, the parameter λ , and the two adversarial losses. Why is cycle consistency important when training with **unpaired images**?

Variants of VAE

Q1. How does a **Convolutional VAE** address the curse of dimensionality? Illustrate the architecture of a Convolutional VAE.

Q2. Explain the architecture and objective of a **Conditional VAE (CVAE)**. How does conditioning on labelled information influence the latent representation and generated output?

Q3. Explain the working principle of **β -VAE**. Derive/write its modified objective function and explain the role of the hyperparameter β in controlling the information capacity of the latent channel. Discuss how β -VAE encourages disentangled representations.

Q4. Explain the motivation for **Ladder VAE (LVAE)**. How does it learn a **hierarchical latent representation**? Compare a conventional hierarchical VAE with the ladder architecture using a suitable diagram.

Q5. A β -VAE has a reconstruction loss of **18.5** and a KL-divergence loss of **2.4** for a given training sample.

The β -VAE objective is given by

$$\mathcal{L}_{\beta\text{-VAE}} = \mathcal{L}_{\text{reconstruction}} - \beta D_{KL}(q(z|x) || p(z)).$$

Calculate the objective value for:

(a) $\beta=1$

(b) $\beta=2$

(c) $\beta=4$

Explain how increasing β changes the relative importance of the KL-divergence term and discuss its effect on the information capacity of the latent representation.

Q6. A vanilla VAE uses a fully connected layer to transform an image representation of size $64 \times 64 \times 3$ into a feature vector of size **512**.

(a) Calculate the number of weights required in this fully connected layer, ignoring biases.

(b) Suppose the convolutional VAE first reduces the image to a feature representation of size $8 \times 8 \times 64$ and then uses a fully connected layer to produce the same 512-dimensional representation. Calculate the number of weights in this layer.

(c) Calculate the reduction in the number of weights and explain why convolutional architectures are preferable for high-dimensional image inputs in VAEs.

References:

1. Class lectures
2. [W1_L1: Course outline deep generative models](#)

